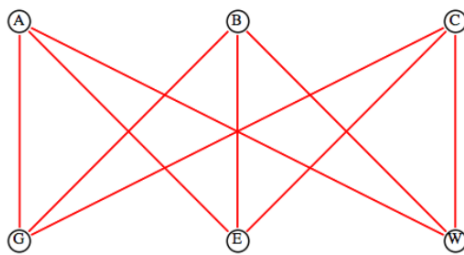


Society, Environment and Engineering Ethics

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Ethics in Artificial Intelligence



Final Report

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Abstract

Artificial Intelligence is becoming more common and part of our daily life. They are changing how we receive health care, education, manage finances and more. These technologies offer exciting benefits but on the other hand they also raise serious ethical concerns. In this paper, we are going to explore the important principles that would guide the use of AI and ML with fairness, accountability, privacy, safety and human oversight. We explore real life cases, where ethical problems occurred such as discrimination in facial recognition, automated decision making etc. We also look at some recent research to understand what is being done and where should we need to improve. As students of computer science and engineering and considering the ethics, we believe that it is essential to make AI not just smarter but also more fair and responsible and we have to make that improvement very soon.

Introduction

Background of the topic

Artificial Intelligence (AI) is a part of computer science. It is focused on creating systems that can perform tasks typically requiring human intelligence. These systems can do problem solving, language understanding, visual perception and decision making. The goal of Artificial Intelligence is to create systems that can think and act like humans.

Machine Learning (ML) is a subfield of AI. This system helps machines to learn from data without being explicitly programmed. ML systems do not follow fixed rules instead these systems identify patterns and improve their performance over time through experience. ML systems are commonly used in applications such as recommendation systems, speech recognition, fraud detection, autonomous vehicles, and medical diagnosis.

Problem Statement

Artificial Intelligence (AI) and Machine Learning (ML) are the latest revolutionary technologies that have taken the world at a new pace. We are using these technologies in various sectors. For example healthcare, finance, education, transportation etc. We can solve tough problems by taking help from AI. But these systems are not just solving complex problems and enhancing productivity, these are also raising some serious ethical concerns. As AI systems are becoming more autonomous and capable, we need to ensure that these systems never become harmful for us.

Objectives of the Topic

The primary objectives of this report are to:

- Explore the foundational ethical principles that should guide the development and deployment of AI, including fairness, accountability, privacy, and safety.
- Analyze real-world case studies to understand how ethical problems, such as bias and lack of safety, manifest in AI systems.
- Examine the current state of AI ethics and identify areas that require significant improvement and further research.

Motivations

The reason behind choosing this topic is that AI and machine learning have become a big part of our daily lives. We are using these smart technologies in our daily lives such as education, healthcare, banks, research, finance, marketing, transportation, e-commerce etc. These are helping us to make decisions that can affect people's health, money, education, and freedom. It is very important for us to use these systems in a fair and safe way. If an AI system makes a mistake or if it is not designed properly, it can hurt

people or treat them unfairly. That's why we must be careful about the rules and values we use when creating and using these technologies. We must use AI and ML in an ethical way. It is our responsibility to stop them from being harmful for us. Our goal is to make sure these powerful tools are used to help people, not to hurt them. If we do this right, AI can make the future better and easier for everyone

Core Ethical Principles in AI

Case Studies

In this section, we explore three prominent case studies that highlight critical ethical issues in the field of AI and Machine Learning.

Amazon's Recruitment Tool: Biased Outcomes while Hiring

Case Study: In 2014, Amazon began to develop AI to automate their hiring process by showing AI resumes and assigning star ratings to candidates [1]. However, in 2015, the company identified a flaw in the system. The AI was not gender-neutral and was showing a bias against women. The algorithm was programmed on old data from a time when the industry was predominated by males. The algorithm learned that males were more suitable for the job and likely more profitable for the company. It penalized female candidates because their resumes contained words like "Women's Debating Club champion" and restricted resumes from women's college graduates [2]. After countless modifications, Amazon could not confirm that the algorithm would work perfectly, so the program was closed.

The Ethical Problem: This case shows a foundational example of algorithmic bias resulting from unrepresentative training data. It demonstrates that a machine can learn social biases from data that are not gender-neutral in the industry. The failure of this algorithm indicates

the massive challenge of creating a fair system and the ethical imperative to audit resources and models for bias before deployment [3].

COMPAS Recidivism Algorithm: Bias in Criminal Justice System

Case Study: COMPAS, an acronym for "Correctional Offender Management Profiling for Alternative Sanctions," is a tool used for predicting the liability of a defendant committing a crime again. US court judges have used this program's output to inform bail and sentencing decisions. In 2016, the investigative journal ProPublica revealed that the system was biased. Their analysis showed that African-American defendants were nearly twice as likely as white defendants to be incorrectly marked for a higher threat of reoffending. Conversely, white defendants were more likely to be mislabeled as low-threat when they did have a criminal history indicating a higher probability of reoffending.

The Ethical Problem: The COMPAS case raises numerous important ethical questions about fairness and transparency in high-stakes public sector algorithms. The tool was not transparent, which is contrary to how a justice system should operate. While it was intended to deliver fairness, the data it was trained on was biased. This contributed to harsher legal consequences for African-American individuals, perpetuating racial inequality within the justice system, even though race was not an explicit input factor [4].

Uber's Self-Driving Car Death: Public Safety and Risky Testing

Case Study: In 2018, an autonomous test car operated by Uber killed a pedestrian in the US. This was the first case of a car driven by AI killing a pedestrian. An alarming fact was that a safety driver was present in the car to ensure the car's course, but they were distracted and failed to observe the car's route. A later investigation found that the car did, in

fact, recognize a human in front but failed to brake, and there was no safety measure in the car to notify the driver that something other than the road was in front.

Ethical Problem: This case study exposes the complex ethics of safety, accountability, and societal experimentation [5]. It raised alarming questions about when AI technology is safe enough to be tested on public roads. The main question is who is responsible for this incident: the engineers or the corporation. Both parties can be held accountable for their lack of ethics concerning public safety.

Future Trends and Challenges

Conclusion

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